

Time Period	Variable	Label	Values	Definition	Code Lists Group	Additional Notes	Date Modified (Initials + Date)	QC F
12	FU_PD MAP_START_DT	The start date for this [MED] MAP	date	For the first MAP in a MMA [MED] dataset (MAP_CNT=1): min(DATE_SERVICE) among [MED] claims For subsequent MAPs in a MMA [MED] dataset (MAP_CNT > 1): min(DATE_SERVICE) among [MED] claims where DATE_SERVICE >= MAP_END_DT of the previous MAP (MAP_CNT-1)		See Figures 1-3 in Section B for more details. MAP_CNT is defined below.		
13	FU_PD MAP_RX_RUNOUT_DT	The [MED] MAP's pharmacy claim runout date.	date	For all [MED] claims with DATE_SERVICE >= MAP_START_DT: if CLAIM_TYPE = pharmacy then: if earliest CLAIM_TYPE = pharmacy claim then: pushout = 0 MAP_RX_RUNOUT_DT=DATE_SERVICE+DAY_SUPPLY-1 if not earliest CLAIM_TYPE = pharmacy claim then: if DATE_SERVICE > max(MAP_RX_RUNOUT_DT, MAP_MED_RUNOUT_DT) then: stop processing this MAP's MAP_RX_RUNOUT_DT and MAP_MEDICAL_RUNOUT_DT else if DATE_SERVICE <= previous MAP_RX_RUNOUT_DT then: pushout = previous MAP_RX_RUNOUT_DT - DATE_SERVICE+1 MAP_RX_RUNOUT_DT = DATE_SERVICE+DAY_SUPPLY-1 + pushout else if DATE_SERVICE > previous MAP_RX_RUNOUT_DT and <= MAP_MED_RUNOUT_DT then: MAP_RX_RUNOUT_DT = DATE_SERVICE + DAY_SUPPLY-1 For all [MED] claims with DATE_SERVICE >=		This variable must be defined concurrently with MAP_MED_RUNOUT_DT while processing a given MAP. For a given [MED], the MAP algorithm uses a patient's DATE_SERVICE, DAYS_SUPPLY, and CLAIM_TYPE from the MMA [MED] dataset to create the patient's MAP(s). The MAP algorithm iterates over a MMA [MED]'s claim/claim-line's DATE_SERVICE and applies pushout and runout rules. If a patient has concurrent pharmacy and medical claims/claim-lines, then pharmacy and medical runout dates will be separately calculated. See Figures 1-3 in Section B for more details.		

A	B	C	D	E	F	G
FU_PD	MAP_CNT	The patient specific MAP order number for [MED]	numeric	=1 for a patient's first [MED] MAP, =2 for a patient's second [MED] second MAP, =3 for a patient's third [MED] MAP, ... and so on.		[MED] Claim level to define, multiple patient-map rows for final MAP_[MED] dataset
FU_PD	MAP_MED_TYPE	Medication abbreviation for this [MED] MAP	E.g., BORT, LEVA, DEXA, etc.	=MED_ABBR	Tab 40	See tab CL MMA ROLLUP for potential values of MED_ABBR.
FU_PD	MAP_MED_CLASS	Class of Medication for this [MED] MAP	E.g., PROTINHB, IMMUNOMOD, MUSTARD, etc.	=MED_CLASS	Tab 40	See tab CL MMA ROLLUP for potential values of MED_CLASS.
FU_PD	MAP_DISCON_FLG	Medication discontinues for at least 90 days after this MAP ends.	0,1	if this MAP not last PATIENT-MAP for this patient and the patient's next MAP's MAP_START_dt - this MAP's MAP_END_DT >= 90 then = 1 else if it is the last PATIENT-MAP for this patient and ENDDATE - this MAP's MAP_END_DT >= 90 then = 1 else = 0		Indicates whether at least 90 days elapse between this MAP's MAP_END_DT and the next MAP's MAP_START_DT

Section B: MAP Algorithm Examples

Examples

Figure 1. Simplified Creation of MAP_JMED)

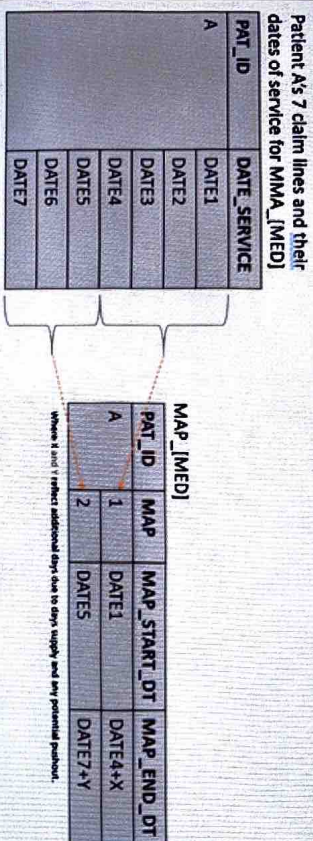
Patient As 7 claim lines and their

Additional Notes

Date Modified
(Initials + Date)

Dr Rev
(Initials + Date)

Figure 1. Simplified Creation of MAP_[MED]



In this simplified example, Patient A's claims are processed one at a time and the first 4 claim/claim-line's day supply are combined to create MAP 1 because their fifth claim/claim-line occurs after the first 4 claims medications (and their day supply/pushout) have run out. Thus, MAP1 end date is set based on the day supply / pushout after combining the first 4 claim lines and the final 3 claim/claim-lines are combined to create MAP 2 (where DATE5 > DATE4+X). Note that while the MAP_START_DT equals an exact date from a claim/claim-line, the MAP_END_DT is an initially unknown value and must be calculated using each claim/claim-line's day supply.

Figure 2. Pharmacy Claim Pushout / Runout Example

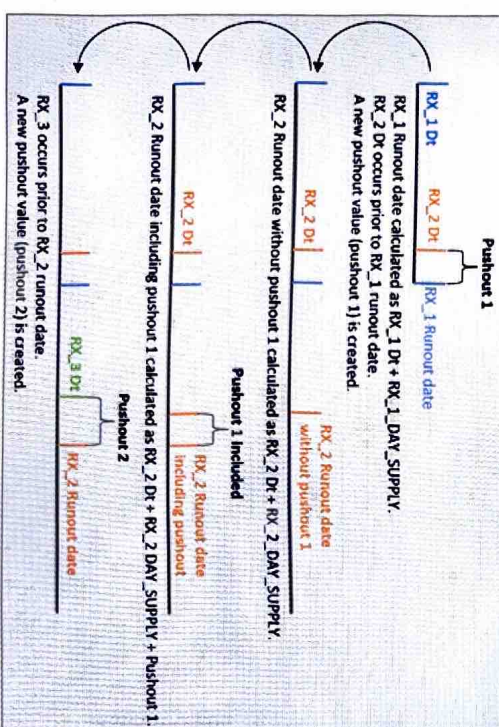


Figure 3. MAP Algorithm Logic Example

MMA_[MED] Dataset Input				MAP Algorithm Details	
PAT_ID	DATE_SERVICE	CLAIM_TYPE	DAY_SUPPLY	Iteration	Pharmacy Runout

Each MAP:

- Has a start and end date.
- Is at the Patient-MAP level (that is, a patient may have multiple MAPs for the same medication and a patient who received multiple MM medications will have at least one MAP for each MM medication received).
- Has nonoverlapping MAP time periods if the a patient has multiple MAPs for a given [MED].
- Is a patient-claim level dataset containing the processed claims having an [MED] MMA medication code belonging to the patients found in the patient input dataset (see INPUT DATASETS tab). For example, MMA_BORT would only contain those processed claims with an NDC or HCPCS code for BORT belonging to patients found in the PATIENT_INPUT dataset.
- A single patient will appear more than once if they have multiple claim/claim-lines indicating use of [MED].

1. Pushout - If a pharmacy claim occurs on day X and has a 28 day supply, assume patient has medication available on day X thru the claim's DATE_SERVICE + 27 days (X+28-1). If a subsequent pharmacy claim for the same medication occurs prior to X+27 on day Y, then pushout_Y is defined as (X+27 - Y) + 1 and captures the days of medication the patient still had available from their earlier fill upon receiving their next pharmacy fill. Conceptually, if a patient has an "early refill" in pharmacy claims, we assume the patient begins the "early refill" [MED] after the runout date of the previous [MED] fill.

2. Pharmacy runout date - The date a patient no longer has medication available based on pharmacy claims. Calculated as the date of the pharmacy claim + days supply + potential pushout.

3. Medical runout date - The date a patient no longer has medication available in their body based on medical claims. Calculated as the date of the medical claim + 28. Pushout is not implemented.

A patient with multiple pharmacy claims, each occurring prior to the current pharmacy runout, will generate a new pushout value for each combination of adjacent claims. See Figure 2 for more details.

For a given patient in MMA_[MED], their earliest claim/claim-line's DATE_SERVICE that occurs after their index date will be the MAP_START_DT of their first [MED] MAP. Set the Pharmacy and

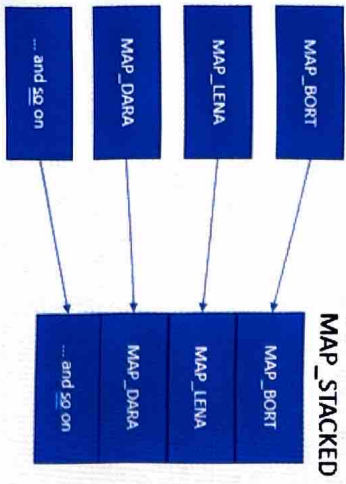
Figure 3. MAP Algorithm Logic Example

MMA_[ME] Dataset Input				MAP Algorithm Details		
PAT_ID	DATE_SERVICE	CLAIM_TYPE	DAY_SUPPLY	iteration	Pharmacy Runout	Medical Runout
A	May 15	Pharmacy	30	1	Jun 13	.
	Jun 10	Pharmacy	30	2	Jul 13	.
	Jun 25	Medical	30	3	Jul 13	Jul 24
	Jul 15	Pharmacy	30	4	Aug 13	Jul 24
	Aug 10	Medical	30	5	Aug 13	Sep 8
	Sep 5	Medical	30	6	Aug 13	Oct 4
	Next claim on NOV 7 occurs after both pharmacy and medical runout dates.					
	Nov 7	Pharmacy	30	7	DEC 6	.

[MED] MAP Algorithm Output for PAT_ID A

PAT_ID	MAP_CNT	MAP_START_DT	MAP_END_DT	MAP_MED_TYPE	MAP_MED_CLASS
A	1	May 15	Oct 4	[MED]	[MED Class]
	2	Nov 7	Dec 6	[MED]	[MED Class]

Figure: stacking MAP_MED datasets together to create MAP_STACKED



Each MAP_[MED] dataset (see tab MAP_MED) is at the patient-map level (a patient can have multiple MAPs for the same MED). After stacking the MAP datasets together, the resulting MAP_STACKED dataset can be thought of as at the MED-patient-map level.

For a given patient in `MMA_[MED]`, their earliest claim/claim-line's `DATE_SERVICE` that occurs after their index date will be the `MAP_START_DT` of their first `[MED]` `MAP`. Set the pharmacy and medical runout dates based on the claim's `CLAIM_TYPE`.

if the chronological next claim/claim-line occurred after both the patient's pharmacy and medical runoff dates, then the set the MAP_END_DT to the maximum of either the pharmacy or medical runoff dates and the claim's DATE_SERVICE will be the MAP_START_DT of the patient's next MAP.

Otherwise, if the next claim occurred on or before either the pharmacy or medical runout dates, use the `CLAIM_TYPE` to update the appropriate runout date (pharmacy or medical), incorporating pushout when updating the pharmacy runout date. Note that it may be possible for a pharmacy claim to occur after the pharmacy runout date but before the medical runout date. This does not end the MAP, rather the pharmacy runout is updated to be the `DATE_SERVICE` of the pharmacy claim + `DAY_SUPPLY-1` (see [Figure 3, MAP Algorithm Iteration 4](#)).

Stack (not merge) each of the MAP_[MED] datasets into one combined dataset MAP_STACKED. MAPS_STACKED will have the exact same variables as the MAP_[MED] datasets but will be comprised of each the MAP_[MED] datasets. If we were to focus on a given patient that was on BORT and LENA, then MAP_STACKED would contain all of the patient's BORT and LENA MAPs (assuming the patient was present in each of the individual MAP_BORT and MAP_LENA datasets).